INFOSYS SE PSEUDOCODE VAULT • ALGORITHMIC COMPLEXITY & THEORY

Time Complexity, OOP Pseudocode, Scoping, Static Variables • Total Solved Questions: 18

[Campus Monk Day 04 Mock Test]

Topic: *Campus Monk Mock Pseudocode*

Q1. What will be the output of the following pseudocode?

```
1.     #include<iostream>
using namespace std;
class Test  {
static  int x;
public:
Test()   {    x++;    }
static int getX()  {   return x;    }   };
int Test::x=0;
int main()   {
cout<< Test::getX()<<" ";
Test t[5];

cout<< Test::getX();      }
```

(A) 0 0
(B) 5 5
(C) 0 5
(D) a) Run time error

Correct Answer: Option (C) : 0 5

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (0 5).

[Campus Monk Day 04 Mock Test]

Topic: *Campus Monk Mock Pseudocode*

Q2. What will be the output of the following pseudocode?

```
1.     #include<iostream>
using namespace std;
class Point    {
Point()
 {  cout<<"Constructor called";   }
};
int main(){
Point t1;
return 0;
}
```

(A) Constructor called
(B) Compile time error
(C) Run time error
(D) Blank line

Correct Answer: Option (B) : Compile time error

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (Compile time error).

[Campus Monk Day 04 Mock Test]

Topic: *Campus Monk Mock Pseudocode*

Q3. What will be the output of the following pseudocode?

```
1.     #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
 {
 int z = x & (x >> 1);
 if (z)
 printf("%d ", x);
 }

 }
```

(A) 7 6 3
(B) 7 5 4
(C) 6 7 8
(D) 6 5 4

Correct Answer: Option (A) : 7 6 3

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (7 6 3).

[Campus Monk Day 04 Mock Test]

Topic: *Campus Monk Mock Pseudocode*

Q4. 1. What will be the output of this statement?

```
printf ("%d", 9/5);
```

(A) 1.8
(B) 1.0
(C) 2.0
(D) None of these

Correct Answer: Option (D) : None of these

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (None of these).

[Campus Monk Day 04 Mock Test]

Topic: *Campus Monk Mock Pseudocode*

Q5. 1. What will be the output of this program?

```
#include<stdio.h>
      int main() {
      int i = 5;
      printf("%d", i = ++i == 6);
      return 0;
      }
```

(A) 2
(B) 6
(C) 4
(D) 1

Correct Answer: Option (D) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (1).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q6. 1. What will be the output of this program?

```
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf("%d", h);
   return 0;
}
```

(A) 1
(B) error: redeclaration of an enumerator
(C) h
(D) 3

Correct Answer: Option (B) : error: redeclaration of an enumerator

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (error: redeclaration of an enumerator).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q7. 1. What will be the output of this program?

```
main ()
{
 if (5 < '5')
        printf("5")
 else
        printf(" NOT EQUAL TO 5.")

}
```

(A) ENQ
(B) 5
(C) I
(D) Not equal to 5

Correct Answer: Option (B) : 5

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q8. What will be the output of the following pseudocode?

```
1.     The value of j at the end of the execution of the
following pseudo code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)

end for
```

(A) 10
(B) 4
(C) 6
(D) 7

Correct Answer: Option (A) : 10

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (A) (10).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q9. What will be the output of the following pseudocode?

```
1.     The value returned by the following function for
f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);

}
```

(A) 5
(B) 6
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (7).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q10. What will be the output of the following pseudocode?

```
Find the output of the following code. #include <stdio.h>
int main(void) { int a = 10, x; x = a-- + +445; printf
("%d", x); return 0; }
```

(A) 19
(B) 435
(C) 455
(D) 23

Correct Answer: Option (C) : 455

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (455).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q11. What will be the output of the following pseudocode?

```
Find the output of the following code. #include <stdio.h>
void main() { int a = 10; if (a == 010) printf("BCA");
else printf("B.Tech"); }
```

(A) BCA
(B) B.tech
(C) 10
(D) Error

Correct Answer: Option (B) : B.tech

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (B.tech).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q12. What will be the output of the following pseudocode?

```
Find the output of the following code. #include <stdio.h>
void main () { int x = 91; x >>= 2; printf ("%d\n", x); }
```

(A) 182
(B) 91
(C) 22
(D) 82

Correct Answer: Option (C) : 22

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (22).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q13. What will be the output of the following pseudocode?

```
What is the output of the following C code? #include
<stdio.h> int main() { int num = 987; int rem; while(num
!= 0) { rem = num % 4; num = num / 10; } printf("%d",
rem); }
```

(A) 0
(B) 1
(C) 10
(D) 5

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (1).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q14. What will be the output of the following pseudocode?

```
What is the output of the following C code? #include
<stdio.h> int main() { float x = 0.0; long int y = 10;
printf("%d", sizeof(y) == sizeof(x + y)); return 0; }
```

(A) 3
(B) 1
(C) 5
(D) 2

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q15. What will be the output of the following pseudocode?

```
What is the output of the following C code? int f(int x,
int *py, int **ppz) { int y, z; **ppz += 1; z = **ppz;
*py += 2; y = *py; x += 3; return x + y + z; } void
main() { int c, *b, **a; c = 4; b = &c; a = &b;
printf("%d ", f(c, b, a)); return 0; }
```

(A) 18
(B) 19
(C) 21
(D) 22

Correct Answer: Option (B) : 19

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (19).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q16. What will be the output of the following pseudocode?

```
What is the output of the following C code? #include
<stdio.h> int main() { int a = 12; void *ptr = (int *)&a;
printf("%d", *ptr); getchar(); return 0; }
```

(A) 12
(B) Compiler Error
(C) Run Time Error
(D) 0

Correct Answer: Option (B) : Compiler Error

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (Compiler Error).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q17. What will be the output of the following pseudocode?

```
Q2.) #include<iostream> using namespace std; int main() {
int a = 3, b = 2, c = 5; int result = (a++ << 2) + (--b
<< 1) - (c-- >> 1) + (a & b); cout << result; return 0; }
```

(A) 12
(B) 14
(C) 10
(D) 16

Correct Answer: Option (A) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (12).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q18. What will be the output of the following pseudocode?

```
int main() { int x = 9, y = 2, z = 6; int a = x & y | z;
printf("%d", a); return 0; }
```

(A) Error
(B) 2
(C) 3
(D) 6

Correct Answer: Option (D) : 6

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (6).